

Abstract No.

Investigating Synergistic Temozolomide Based Combinations with Rifampicin, Ulipristal, and EdU in U87 Glioblastoma Cells

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Glioblastoma (GBM) is the most aggressive primary malignant brain tumor in adults and remains associated with poor survival despite current standard therapy. The clinical efficacy of temozolomide (TMZ), the standard chemotherapeutic agent for GBM, is limited by intrinsic and acquired resistance mechanisms, including enhanced DNA damage repair. Drug repurposing based combination strategies may improve treatment response by simultaneously targeting complementary resistance-associated pathways. This study aimed to evaluate the in vitro cytotoxic efficacy and synergistic potential of TMZ combined with three repurposed agents targeting distinct pathways: rifampicin (an RNA polymerase inhibitor), ulipristal (a progesterone receptor antagonist), and 5-ethynyl-2-deoxyuridine (EdU; a DNA damage enhancer), using the U87 human glioblastoma cell line.

IC₅₀ values were determined by MTT cell viability assays following dose-response experiments performed within the following concentration ranges: TMZ (100 µM–1 mM), EdU (1–50 µM), ulipristal (100 µM–1 mM), and rifampicin (50–250 µM). Based on monotherapy IC₅₀ values, eight treatment groups were established: control, TMZ, TMZ+EdU, TMZ+rifampicin, TMZ+ulipristal, TMZ+EdU+rifampicin, TMZ+EdU+ulipristal, and TMZ+EdU+ulipristal+rifampicin. Drug interactions were quantified by Combination Index (CI) analysis using the Chou–Talalay method implemented in CompuSyn.

Monotherapy assays enabled determination of IC₅₀ values for each agent. In combination analyses, multiple TMZ based regimens showed synergistic cytotoxicity (CI < 1) and greater anti-proliferative effects than single-agent treatment in U87 cells. These findings establish an in vitro basis for subsequent analyses of real-time cytotoxicity kinetics, apoptosis induction, cell invasion, and resistance-associated metabolic alterations by untargeted LC-MS/MS metabolomics. Overall, the results support the potential of TMZ based multi-drug combinations as a preclinical strategy for targeting resistance associated pathways in glioblastoma.

Keywords: Glioblastoma, Temozolomide, EdU, Ulipristal, Rifampicin, Combinational therapy.



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